

SOUTH BASS ISLAND, OH, USA

WMO: 994340

Lat: 41.6289N

Lon: 82.8413W

Elev: 177

StdP: 99.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.1	(c) -12.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 18.9	(k) -3.6	(l) 17.4	(m) -4.0	(n) 7.5	(o) 230	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.4	(c) 29.4	(d) N/A	(e) 28.2	(f) N/A	(g) 27.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.4	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.7	(b) 15.2	(c) 13.2	(e) -17.0	(f) 32.1	(g) 3.2	(h) 1.7	(i) -19.3	(j) 33.4	(k) -21.2	(l) 34.3	(m) -23.0	(n) 35.3	(o) -25.3	(p) 36.5		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-3.2	-2.8	2.0	8.7	14.9	20.9	23.7	23.2	19.8	13.1	6.3	0.6
	DBStd	10.44	5.59	5.22	4.85	4.26	4.05	3.12	2.49	2.31	3.25	3.96	4.44	4.42
	HDD10.0	1527	408	358	253	73	4	0	0	0	14	126	291	
	HDD18.3	3288	667	591	507	288	122	11	1	22	169	361	548	
	CDD10.0	1775	0	0	4	35	157	326	425	410	293	109	14	1
	CDD18.3	494	0	0	0	1	16	87	167	152	64	7	0	0
	CDH23.3	2239	0	0	0	1	46	313	956	722	189	12	0	0
	CDH26.7	322	0	0	0	0	7	43	159	89	24	1	0	0
Wind	WSAvg	6.1	7.6	7.2	6.5	7.0	5.7	4.7	4.4	4.3	4.9	6.5	7.3	7.4
Precipitation	PrecAvg	956	64	55	67	92	94	100	95	88	90	70	72	71
	PrecMax	1162	142	135	138	137	183	159	218	215	206	119	139	184
	PrecMin	675	19	14	28	51	31	34	41	20	31	19	21	31
	PrecStd	125	31	30	27	25	43	35	40	45	40	24	26	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	11.4	17.4	22.4	27.5	29.7	31.3	30.5	29.4	25.0	17.9	13.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	9.7	7.9	14.0	19.7	24.3	27.9	29.6	28.8	26.9	22.4	15.6	10.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	7.0	5.5	11.4	17.6	22.5	26.5	28.4	27.6	25.3	20.5	14.0	8.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	3.9	3.2	8.6	15.3	20.7	25.1	27.3	26.6	24.2	18.7	12.5	6.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.7	5.8	6.3	5.8	5.5	5.4	5.0	5.2	5.1	4.8	4.3
		MCDBR	8.1	8.0	9.7	10.0	7.9	6.9	6.4	5.8	6.6	7.1	7.3	6.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.309	0.333	0.359	0.400	0.436	0.448	0.442	0.433	0.387	0.365	0.326	0.316
	taud		2.394	2.305	2.314	2.251	2.192	2.208	2.245	2.257	2.362	2.403	2.475	2.419
	Ebn at Noon		837	866	886	870	844	832	832	829	846	822	814	795
	Edh at Noon		81	103	115	132	143	142	135	130	109	93	75	73
All-Sky Solar Radiation	RadAvg		1.44	2.28	3.44	4.60	5.43	6.05	6.08	5.33	4.30	2.81	1.77	1.20
	RadStd		0.15	0.31	0.31	0.36	0.46	0.40	0.34	0.20	0.36	0.22	0.25	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	N/A

Nomenclature: See separate page